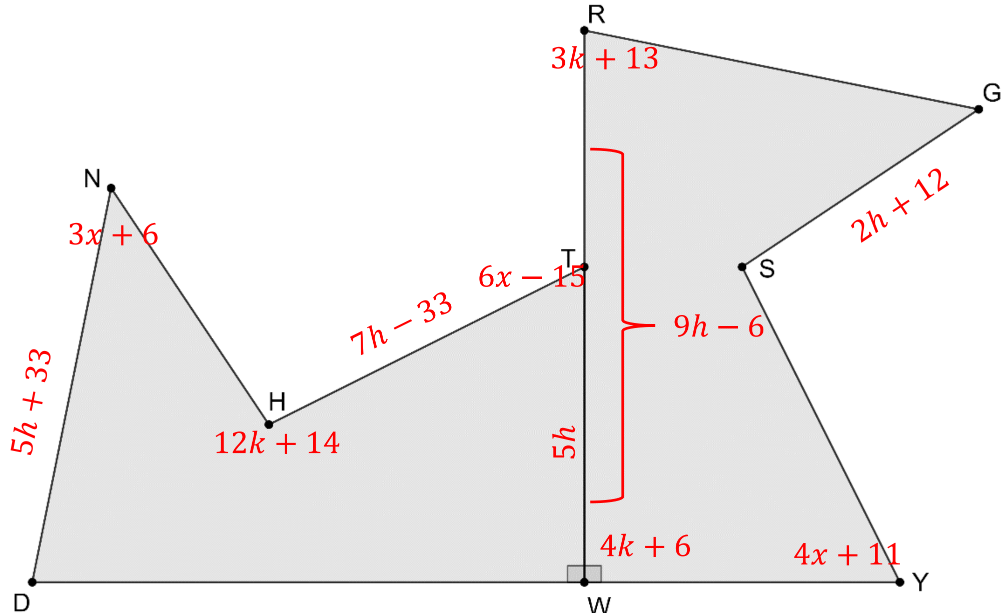
Geometry
Unit 5
Lesson 9 Practice

Name___Answer Key_______
Date______________________
Period_____________________

Congruent polygons $DNHTW$ and $RGSYW$ are shown, where $\angle DWT$ and $\angle RWY$ are right angles.



Some of the angle measures of $DNHTW$ and $RGSYW$ are shown.

$m\angle HTW = (6x - 15)°$ $\quad$ $m\angle NHT = (12k + 14)°$ $\quad$ $m\angle DNH = (3x + 6)°$

$m\angle WRG = (3k + 13)°$ $\quad$ $m\angle WYS = (4x + 11)°$ $\quad$ $m\angle YWR = (4k + 6)°$

1. Determine the value of k.

$4k + 6 = 90$
$4k = 84$
$k = 21$

2. Determine the value of x.

$6x - 15 = 4x + 11$
$2x = 26$
$x = 13$

Determine each angle measure.

3. $m\angle GSY =$ <u>266°</u>

$m\angle GSY = m\angle NHT$
$= 12(21) + 14$
$= 266°$

4. $m\angle RGS =$ <u>45°</u>

$m\angle RGS = m\angle DNH$
$= 3(13) + 6$
$= 45°$

5. $m\angle WDN =$ <u>76°</u>

$m\angle WDN = m\angle WRG$
$= 3(21) + 13$
$= 76°$

Some of the measures of the side lengths of $DNHTW$ and $RGSYW$ are shown.

$ND = 5h + 33$ $\quad$ $HT = 7h - 33$ $\quad$ $RW = 9h - 6$

$GS = 2h + 12$ $\quad$ $TW = 5h$ $\quad$ $DW + WY = 330$

6. Determine the value of h.

$9h - 6 + 5h = 330$
$14h - 6 = 330$
$14h = 336$
$h = 24$

7. What is the length of $\overline{HN}$?

$HN = SG$
$= 2(24) + 12$
$= 48 + 12$
$= 60$
$HN = 60$

8. What is the length of $\overline{SY}$?

$SY = HT$
$= 7(24) - 33$
$= 135$
$SY = 135$

9. What is the length of $\overline{RG}$?

$RG = DN$
$= 5(24) + 33$
$= 153$
$RG = 153$

10. What is the length of $\overline{DY}$?

330